HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

Exercises and Worked Examples + Solutions

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'
Chapter 1

Prof. Dr. Christoph Gerhard, 2024

1 Basics of plasma physics

Plasmas in space and nature

‘Tell me and I forget.
Show me and I remember.
Let me do and I understand.’

Confucius

Plasmas in space and nature

???

- **Exercise 1.1**
  - Determine the electric charge within a thundercloud (aka *cumulonimbus*) with an area of A = 1 km² where the electric field between the differently charged upper and lower part of the cloud amounts to $E = 3 \times 10^6$ V/M.

Plasmas in space and nature

! ! !

- **Exercise 1.1 – Solution**

We assume the cloud to be a parallel-plate capacitor where the voltage is

$$U = E \cdot d \quad (1).$$

The capacity of this capacitor is given by

$$C = \varepsilon_0 \cdot A/d \quad (2).$$

Moreover, the electric charge is

$$Q = C \cdot U \quad (3).$$

Inserting (1) and (2) into (3) and cancelling down the distance d gives

$$Q = \varepsilon_0 \cdot A \cdot E \quad (4).$$

The electric charge thus amounts to

$$Q = 8.85 \times 10^{-12}\ \mathrm{As/Vm} \cdot 10^{6}\ \mathrm{m}^2 \cdot 3 \times 10^{6}\ \mathrm{V/m} = \underline{26.6\ \mathrm{C}}.$$

Plasmas in space and nature

???

- **Exercise 1.2**
  - The voltage between the ground and a thundercloud (aka *cumulonimbus*) at an altitude of d = 11.5 amounts to V = 100 MV. From this cloud, a cylindrically-shaped flash with a diameter of 2 inches ignites directly to the ground (directly = straight line). As a mean value, the resistor of air is $R = 1.5 \times 10^{14}\ \Omega$.

a) Determine the electric field E between the ground and the thundercloud.
b) Determine the current density J within the flash.
c) Determine the electrical power P within the flash.

Plasmas in space and nature

! ! !

- **Exercise 1.2 – Solution**

  a)

  $U = E \times d \rightarrow E = U/d = 100\ MV/11.5\ km \approx \underline{8696\ V/m}$

  b)

  $J = I/A$ where

  - $I = U/R = 100\ MV/1.5 \times 10^{14}\ \Omega \approx 6.7 \times 10^{-7}\ A$

    and

  - $A = (D/2)^2 \times \pi \approx 2 \times 10^{-3}\ m^2$

    with D = 2 inches = 5.08 cm

  $\rightarrow J = 6.7 \times 10^{-7}\ A/2 \times 10^{-3}\ m \approx \underline{3.4 \times 10^{-4}\ A/m^2}$

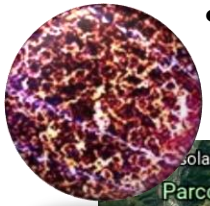
  c)

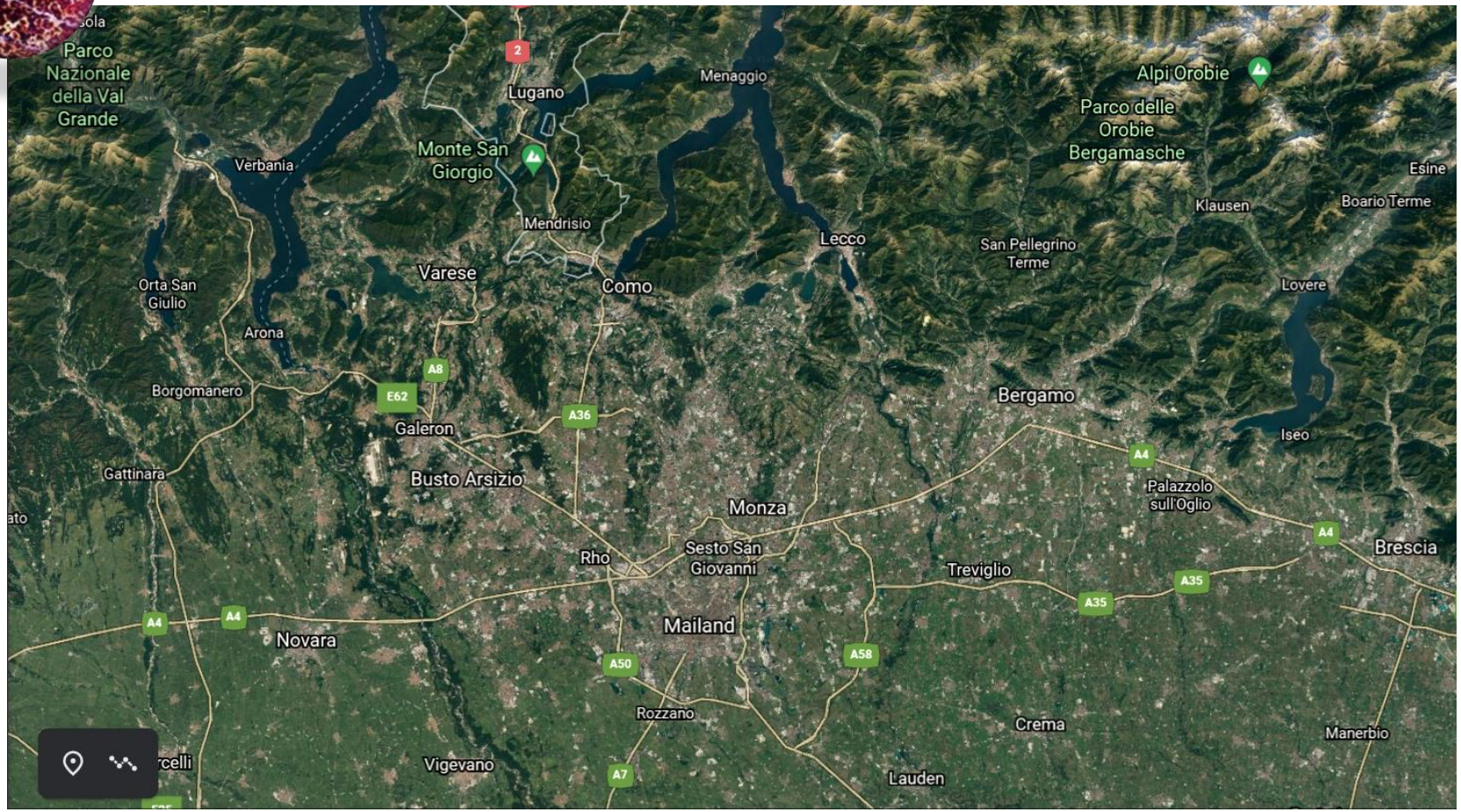
  $P = 100\ MV \times 6.7 \times 10^{-7}\ A \approx \underline{67\ W}$

Plasmas in space and nature

???

- **EXERCISE 1.3**
  - Determine the number of particles within a neutron star with a diameter of 20 km.





Plasmas in space and nature

! ! !

- **EXERCISE 1.3 – Solution**
  - Assuming the neutron star to be a perfect sphere, its volume amounts to

$$V = \frac{4}{3} \cdot r^3 \cdot \pi = \frac{4}{3} \cdot 10.000 m^3 \cdot \pi \approx 4.2 \cdot 10^{12} m^3.$$

  - Since the particle density within a neutron star is $n = 10^{42}$ m^{-3}, the number of particles in the given neutron star is

$$\mathrm{N} = 4.2 \cdot 10^{12} m^3 \cdot 10^{42} m^{-3} = 4.2 \cdot 10^{54}.$$

Plasma parameters and mechanisms

???

- **Exercise 1.4**
  - A hydrogen plasma (i.e., hydrogen in plasma state) in local thermodynamic equilibrium at a temperature of 10^7K expands within vacuum.

  - Determine the expansion velocity of the plasma.

Plasma parameters and mechanisms

! ! !

- **Exercise 1.4 – Solution**

The expansion velocity of the plasma is given by the ion sound velocity,

$$c_s = \sqrt{k_B \cdot \frac{T_e + 3 \cdot T_i}{m_i}}.$$

Since the plasma is in LTE, both the electron and ion temperature are 10^7K. The mass of the hydrogen ion can be determined as follows: The mass of the hydrogen atom is 1,00784 u = $1{,}67356 \cdot 10^{-27}$ kg. Since hydrogen features merely one single electron and can thus only be singly ionised, subtracting the electron mass gives the ion mass:

$$1{,}67356 \cdot 10^{-27} kg - 9{,}10938 \cdot 10^{-31} kg = 1{,}67265 \cdot 10^{-27} kg.$$

The ion sound velocity and plasma expansion velocity finally amounts to

$$c_s = \sqrt{1{,}38065 \cdot 10^{-23} \frac{m^2 \cdot kg}{s^2 \cdot K} \cdot \frac{10^7 K + 3 \cdot 10^7 K}{1{,}67265 \cdot 10^{-27} kg}} = 574.605 \frac{m}{s}.$$

(Footnote: In principle, the electron mass is four orders of magnitudes lower than the atom mass and could thus be neglected.)

Plasma parameters and mechanisms

???

- **Exercise 1.5**
  - The electron temperature and density with a plasma in local thermodynamic equilibrium are 1 eV and $10^{15}/m^3$, respectively.

  - Determine the Debye length of this plasma.

Plasma parameters and mechanisms

! ! !

- **Exercise 1.5 – Solution**

In the present case, the Debye length for electrons corresponds to the one for ions since the plasma is in local thermodynamic equilibrium. The Debye length is thus

$$\lambda_D = \sqrt{\frac{8.85 \cdot 10^{-12} \frac{As}{Vm} \cdot 1.38 \cdot 10^{-23} \frac{m^2 \cdot kg}{s^2 \cdot K} \cdot 11604K}{\frac{10^{15}}{m^3} \cdot (1.6 \cdot 10^{-19} As)^2}} = 2.35 \cdot 10^{-4} \approx 235 \mu m.$$

Notes: $1V = 1 \frac{kg \cdot m^2}{A \cdot s^3}$ and 1 eV = 11604K

Plasma parameters and mechanisms

???

- **Exercise 1.6**
  - We consider an argon plasma where all atoms within the plasma are singly ionised and the electron temperature is 1 eV.

  a) Determine the ratio or factor of the electron velocity and the ion velocity within this plasma.
  b) Determine the potential (given in eV) within the plasma sheath.

Plasma parameters and mechanisms

! ! !

- **Exercise 1.6 – Solution**

a)
The wanted factor is described by the condition

$$\frac{v_e}{v_i} > \sqrt{\frac{m_i}{m_e}},$$

it is thus at least higher than the square root of the masses of the singly ionised argon atom and an electron. The mass of an argon atom is 39,94800 u = $6{,}63400 \cdot 10^{-26}$ kg, the electron mass amounts to $9{,}10938 \cdot 10^{-31}$ kg. The mass of a singly ionised argon atom is thus

$$6{,}63400 \cdot 10^{-26} kg - 9{,}10938 \cdot 10^{-31} kg = 6{,}63391 \cdot 10^{-26} kg.$$

The wanted factor is then

$$\frac{v_e}{v_i} > \sqrt{\frac{6{,}63391 \cdot 10^{-26} kg}{9{,}10938 \cdot 10^{-31} kg}} \approx 270.$$

Plasma parameters and mechanisms

! ! !

- **Exercise 1.6 – Solution**

b)
The potential of the sheath follows from

$$\Phi \approx -T_e \cdot ln \sqrt{\frac{m_i}{2 \cdot \pi \cdot m_e}}$$

and amounts to

$$\Phi \approx -1eV \cdot ln \sqrt{\frac{6{,}63391 \cdot 10^{-26} kg}{2 \cdot \pi \cdot 9{,}10938 \cdot 10^{-31} kg}} = -4{,}67897 eV.$$

Plasma parameters and mechanisms

???

- **Exercise 1.7**
  - An electron with a radius of 2.8×10^{-15} m (note: estimation, since sizes of quantum objects cannot be determined definitely) propagates within a gas that purely consists of atomic nitrogen where the particle density is $n_N = 13 \times 10^{17}/\text{cm}^3$. The mean radius of a nitrogen atom is 65 pm.

  - Determine the mean free length of path of the electron between two collisions.

Plasma parameters and mechanisms

! ! !

- **Exercise 1.7 – Solution**

For the determination of the mean free length of path, the effective cross section needs to be calculated initially. It follows from the radii of the involved particles, i.e. the electron and a nitrogen atom, according to

$$\sigma = \pi \cdot (r_1 + r_2)^2.$$

It amounts to

$$\sigma = \pi \cdot (2.8 \cdot 10^{-15} m + 6.5 \cdot 10^{-11} m)^2 = 1.3 \cdot 10^{-20} m^2.$$

The mean free length of path, given by

$$\lambda = \frac{1}{n \cdot \sigma},$$

is thus

$$\lambda = \frac{1}{\frac{13 \cdot 10^{23}}{m^3} \cdot 1.3^{-20} m^2} \approx 5.9 \cdot 10^{-5} m = 59 \mu m.$$

Plasma parameters and mechanisms

???

- **Exercise 1.8**
  - Calculate the most probable velocity of an electron with in plasma where the electron temperature amounts to ≈ 7736 K.

Plasma parameters and mechanisms

! ! !

- **Exercise 1.8 – Solution**

The most probable velocity of an electron is given by

$$v_{max} = \sqrt{\frac{2 \cdot k_B \cdot T}{m}}.$$

In the present case, i.e. for an electron with the mass $9.10938 \cdot 10^{-31}$ kg, it amounts to

$$v_{max} = \sqrt{\frac{2 \cdot 1.38065 \cdot 10^{-23} \frac{m^2 \cdot kg}{s^2 \cdot K} \cdot 7736K}{9.10938 \cdot 10^{-31} kg}} = 484.251 \frac{m}{s}.$$

Plasma parameters and mechanisms

???

- **Exercise 1.9**
  - Determine the plasma frequency of a technical plasma with an electron density of $10^{15}/m^3$.

Plasma parameters and mechanisms

! ! !

- **Exercise 1.9 – Solution**

The plasma frequency is given by

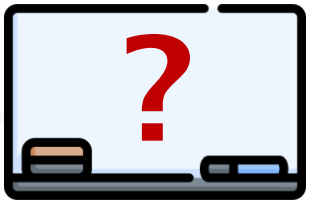
$$\omega_p = \sqrt{\frac{e^2 \cdot n_e}{m_e \cdot \epsilon_0}}$$

and thus amounts to

$$\omega_p = \sqrt{\frac{(1{,}60218 \cdot 10^{-19} As)^2 \cdot \frac{10^{15}}{m^3}}{9{,}10938 \cdot 10^{-31} kg \cdot 8{,}85419 \cdot 10^{-12} \frac{As}{Vm}}} = 1.783.990.256 s^{-1} \approx 1{,}78 GHz$$

where
e = 1,60218·10^{-19} A·s,
m_e = 9,10938·10^{-31} kg und
ε_0 = 8,85419·10^{-12} A·s·V^{-1}·m^{-1}.
Note: $1V = 1 \frac{kg \cdot m^2}{A \cdot s^3}$

Categorisation and types of plasmas



- **Worked example 1.1**
  - A plasma is ignited at a pressure of 1013 hPa. Within this plasma, the electron temperature, the ion temperature as well as the gas temperature is 2 eV. The degree of ionisation amounts to 0.95 and the particle density is 10^{15}.

  - Classify this plasma. Which category/categories is/are appropriate?

Categorisation and types of plasmas

- **Worked example 1.1 – Answer**
  - Classification/categorisation:
    - p =1013 hPa
      → atmospheric pressure plasma
    - T = 2 eV = 23208K ≈ 2.3 x 10^4K
      → low temperature plasma in local thermodynamic equilibrium
    - κ = 0.95 ≈ 1
      → fully ionised plasma
    - T ≈ 2.3 x 10^4K and n = 10^{15}
      → ideal plasma, since here, $N_D = \frac{4\pi}{3}\lambda_D^3 n \gg 1$
    - **Ideal and fully ionised low temperature plasma in local thermodynamic equilibrium at atmospheric pressure**

Categorisation and types of plasmas

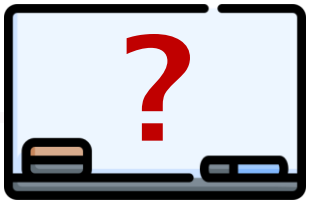
- **Worked example 1.1 – Answer**
  - Classification/categorisation:
  - Condition $N_D = \frac{4\pi}{3}\lambda_D^3 n \gg 1$ fulfilled?
  - Yes, since

$$N_D = \frac{4\pi}{3} \cdot \lambda_D^3 \cdot n = \frac{4\pi}{3} \cdot 3.7^{-11} m^3 \cdot \frac{10^{15}}{m^3} = 154675 \gg 1$$

where

$$\lambda_D = \sqrt{\frac{8.85 \cdot 10^{-12} \frac{As}{Vm} \cdot 1.38 \cdot 10^{-23} \frac{m^2 \cdot kg}{s^2 \cdot K} \cdot 23208K}{\frac{10^{15}}{m^3} \cdot (1.6 \cdot 10^{-19} As)^2}} = 3.33 \cdot m^{-4} \approx 333 \mu m$$

Categorisation and types of plasmas



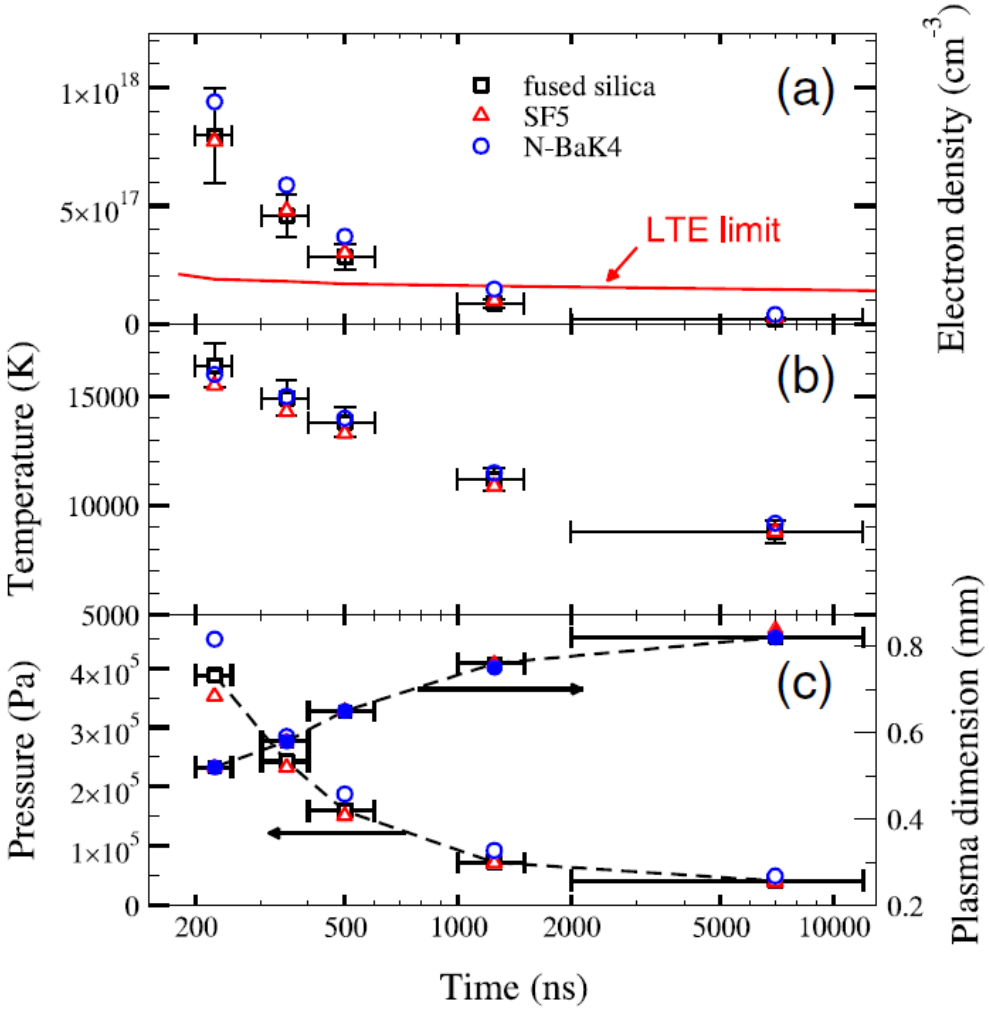
- **Worked example 1.2**
  - The table below shows the time-dependency of different plasma parameters, starting at 225 ns after ignition of the plasma where the base pressure was 500 mbar.

	225 ns	350 ns	500 ns	1250 ns	7 μs
n_e (10^{17} cm^{-3})	8.0	4.7	2.8	0.87	0.2
T (10^3 K)	16.0	14.5	13.4	10.9	8.6
P (10^5 Pa)	4.2	2.7	1.8	0.85	0.55
n_{tot} (10^{17} cm^{-3})	15	11	7.9	5.0	4.5
L (mm)	0.52	0.58	0.65	0.75	0.78

  - Visualise and discuss the given data.

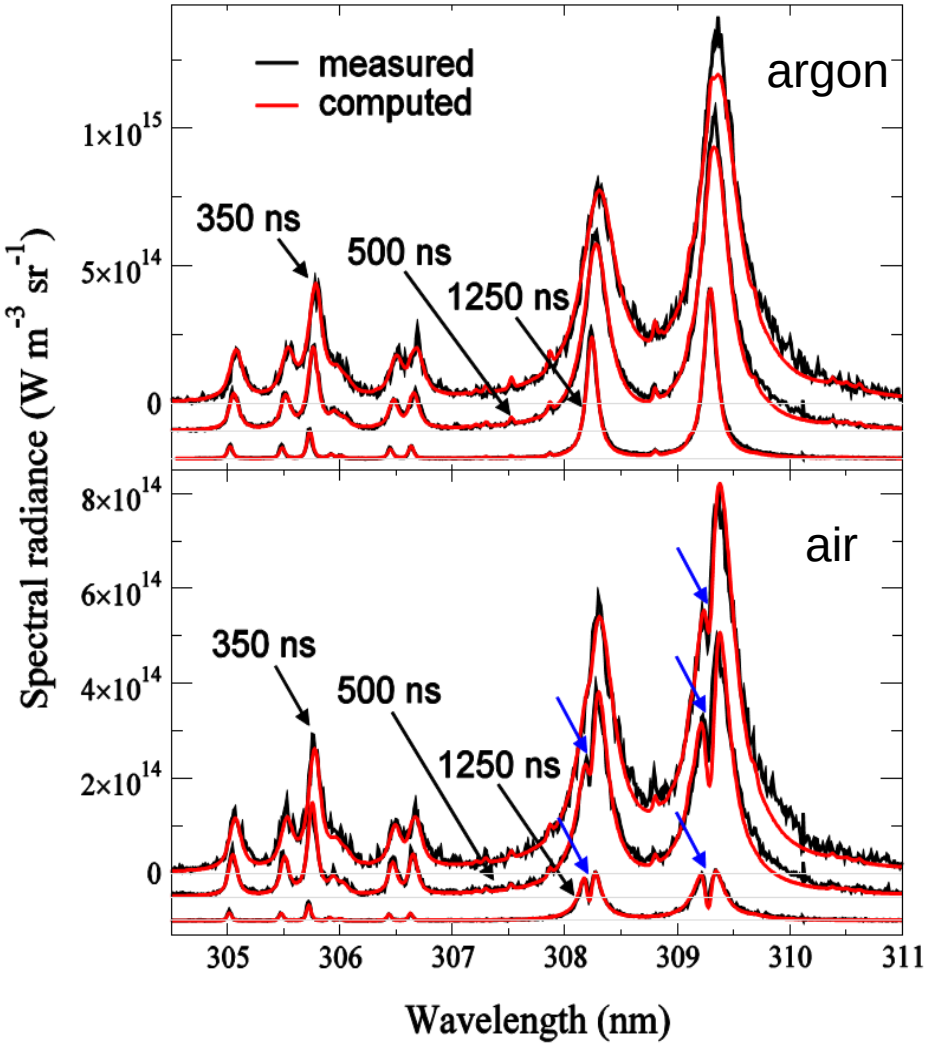
Categorisation and types of plasmas

- **Worked example 1.2 – Answer**
  - Plasma is expanding over time since pressure decreases and length increases
  - Plasma cools down over time since temperature drops
  - Plasma collapses over time due to recombination since electron density decreases



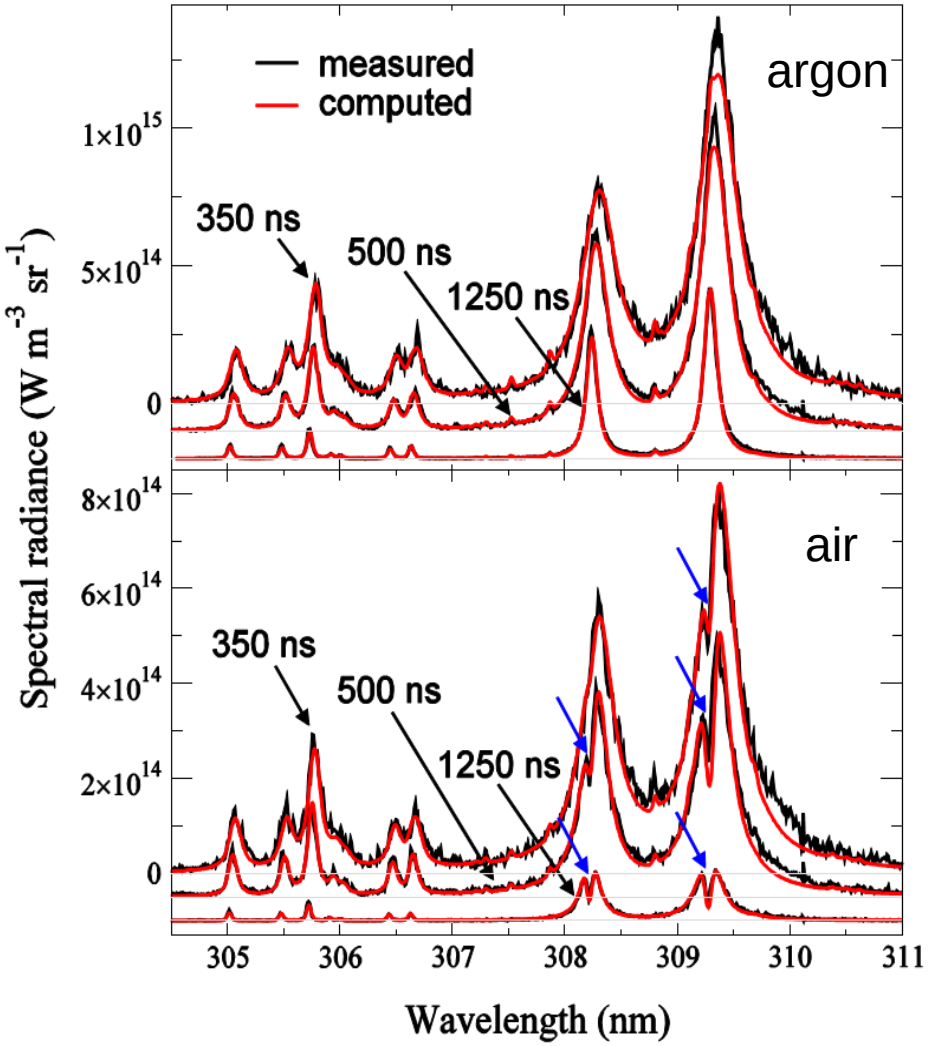
Categorisation and types of plasmas

- **Worked example 1.3**
  - The figure shows the emission spectrum of a plasma ignited in argon and air, respectively.
  - Do you have any approach for explaining the observed differences highlighted by the blue arrows?
    Hint: In air, the electron density of the plasma is much higher than in argon…



Categorisation and types of plasmas

- **Worked example 1.3 – Answer**
  - The observed difference, i.e. absorption dips, are caused by plasma shielding by the dense plasma in air. Here, the plasma frequency,

$$\omega_p = \sqrt{\frac{e^2 n_e}{m_e \epsilon_0}}$$

is is higher than the frequency of the light emitted by the plasma, the plasma thus acts as a mirror for the irradiation emitted by itself.